

IGCSE BIOLOGY

Comprehensive Study Notes

Cell Biology · Human Physiology · Ecology

Includes Examinable Examples & Practice Exercises

Syllabus	Cambridge IGCSE Biology (0610)
Topics Covered	Cell Biology · Human Physiology · Basic Ecology
Difficulty Level	Core & Extended (suitable for all candidates)
Includes	Definitions · Diagrams · Worked Examples · Exam-style Questions

How to use these notes

These notes follow the Cambridge IGCSE Biology syllabus. Each section begins with clear definitions and key concepts, followed by detailed explanations, worked examples drawn from past papers, and exam-style practice questions. Green boxes highlight key terms; orange boxes highlight common exam pitfalls. Work through each section systematically, attempt the questions before checking the answers, and pay close attention to the Exam Tips.

TABLE OF CONTENTS

UNIT 1: CELL BIOLOGY

- 1.1 Cell Structure — Plant vs Animal vs Bacterial
- 1.2 Cell Organisation
- 1.3 Diffusion, Osmosis & Active Transport
- 1.4 Cell Division — Mitosis & Meiosis

UNIT 2: HUMAN PHYSIOLOGY

- 2.1 Nutrition & Enzymes
- 2.2 The Digestive System
- 2.3 The Respiratory System
- 2.4 The Circulatory System
- 2.5 Homeostasis — Thermoregulation & Blood Glucose
- 2.6 The Nervous System & Hormones
- 2.7 Excretion — The Kidneys

UNIT 3: BASIC ECOLOGY

- 3.1 Ecosystems, Habitats & Niches
- 3.2 Food Chains & Food Webs
- 3.3 Energy Flow & Ecological Pyramids
- 3.4 Nutrient Cycles — Carbon & Nitrogen
- 3.5 Population Dynamics
- 3.6 Human Impacts & Conservation

PRACTICE QUESTIONS

- Section A — Short Answer Questions
- Section B — Structured Questions
- Section C — Extended Writing

UNIT 1: CELL BIOLOGY

1.1 Cell Structure

Introduction to Cells

The cell is the basic structural and functional unit of all living organisms. Robert Hooke first described cells in 1665 when examining cork under a microscope. All organisms are composed of one or more cells (Cell Theory).

Cell Theory: All living things are made of cells; the cell is the basic unit of life; all cells come from pre-existing cells

Animal Cell — Organelles & Functions

Organelle	Function
Cell membrane	Controls what enters/leaves the cell (selectively permeable)
Nucleus	Contains DNA; controls cell activities and protein synthesis
Cytoplasm	Jelly-like fluid where chemical reactions occur
Mitochondria	Site of aerobic respiration; produces ATP (energy)
Ribosomes	Site of protein synthesis (translation of mRNA)
Endoplasmic reticulum (ER)	Rough ER: transports proteins. Smooth ER: lipid synthesis
Golgi apparatus	Modifies, packages and secretes proteins
Lysosomes	Contain digestive enzymes; break down waste/pathogens

Plant Cell — Additional Organelles

Additional Structure	Function / Notes
Cell wall	Made of cellulose; provides rigidity and support
Large vacuole	Stores cell sap; maintains turgor pressure
Chloroplasts	Contain chlorophyll; site of photosynthesis
Amyloplasts	Store starch (e.g. in potato cells)

Bacterial Cell

Bacterial cells are **prokaryotic** — they lack a membrane-bound nucleus. Their DNA is a single circular chromosome free in the cytoplasm. Bacteria may also have plasmids (small circular DNA), flagella (for movement), and a cell wall made of peptidoglycan.

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Feature' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9.5, greek=0, italic=0, link=[], rise=0, text='Feature', textColor=Color(.129412,.129412,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Animal Cell' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9.5, greek=0, italic=0, link=[], rise=0, text='Animal Cell', textColor=Color(.129412,.129412,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Plant Cell' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9.5, greek=0, italic=0, link=[], rise=0, text='Plant Cell', textColor=Color(.129412,.129412,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Bacterial Cell' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9.5, greek=0, italic=0, link=[], rise=0, text='Bacterial Cell', textColor=Color(.129412,.129412,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Nucleus	✓ (membrane-bound)	✓ (membrane-bound)	✗ (nucleoid only)
Cell wall	✗	✓ (cellulose)	✓ (peptidoglycan)
Chloroplasts	✗	✓ (in green cells)	✗
Mitochondria	✓	✓	✗
Ribosomes	✓ (80S)	✓ (80S)	✓ (70S, smaller)
Plasmids	✗	✗	✓ (sometimes)
Size	~10–30 µm	~10–100 µm	~1–10 µm

■ **EXAM TIP:** Examiners often ask you to state **TWO** differences between plant and animal cells. Always include: **cell wall, vacuole, and chloroplasts**. Do not say 'nucleus' — animal cells also have a nucleus!

■ WORKED EXAMPLE — Comparing Cell Types (Cambridge-style)

A student observes two cells under a microscope. Cell A has a large central vacuole and a green organelle. Cell B is much smaller, has no nucleus, but has a cell wall. **Identify** each cell and **justify** your answer.

Cell A = Plant cell. Evidence: large central vacuole and chloroplasts (green organelle). Cell B = Bacterial cell. Evidence: no membrane-bound nucleus (prokaryotic) and very small size (~1–10 µm).

1.2 Cell Organisation

Cells are organised into increasingly complex levels:

Level	Description & Example
Cells	Basic units (e.g. red blood cells, muscle cells)
Tissues	Groups of similar cells performing one function (e.g. muscle tissue, epithelial tissue)
Organs	Groups of tissues performing one function (e.g. heart, lungs, stomach)

Organ systems	Groups of organs working together (e.g. circulatory system, digestive system)
Organism	A complete living entity made of organ systems

Specialisation: The process by which a cell develops specific features to carry out a particular function

Example — Specialised Cells:

- **Red blood cells (erythrocytes):** Biconcave disc shape → increases surface area for O_2 absorption. No nucleus → more space for haemoglobin. Thin membrane → O_2 diffuses easily.
- **Root hair cells:** Long extension → huge surface area for water/mineral absorption. Thin cell wall → fast diffusion.
- **Nerve cells (neurones):** Long axon → carries impulses over large distances. Myelin sheath → speeds up transmission.
- **Sperm cells:** Many mitochondria → energy for tail movement. Acrosome (enzyme-filled cap) → penetrates egg.

■ **EXAM TIP:** A common 4-mark exam question: 'Describe how a red blood cell is adapted for its function.' State **FOUR** distinct adaptations with an explanation of how each helps.

1.3 Diffusion, Osmosis & Active Transport

Diffusion

Diffusion: The net movement of molecules or ions from a region of higher concentration to a region of lower concentration, down a concentration gradient, as a result of random movement

Factors affecting rate of diffusion:

- Concentration gradient — greater difference = faster diffusion
- Surface area — larger area = more molecules crossing per second
- Temperature — higher temperature = more kinetic energy = faster diffusion
- Diffusion distance — shorter distance = faster rate
- Molecular size — smaller molecules diffuse faster

Osmosis

Osmosis: The net movement of WATER molecules from a region of higher water potential to a region of lower water potential, through a selectively permeable membrane

Key terminology:

- **Hypotonic solution:** lower solute concentration than the cell → cell gains water → turgid (plants) or may lyse (animals)
- **Hypertonic solution:** higher solute concentration than the cell → cell loses water → plasmolysis (plants) or crenation (animals)
- **Isotonic solution:** same solute concentration as the cell → no net water movement

■ WORKED EXAMPLE — Osmosis Calculation

A potato cylinder of mass 5.0 g is placed in a 1.0 mol/dm³ sucrose solution for 30 minutes. Its final mass is 4.3 g. Calculate the percentage change in mass.

Working: Change in mass = 4.3 – 5.0 = –0.7 g

$$\% \text{ change} = (-0.7 \div 5.0) \times 100 = -14\%$$

The negative value indicates water has LEFT the cell by osmosis (cell is in hypertonic solution).

■ **EXAM TIP:** Always express osmosis results as **PERCENTAGE** change in mass, not absolute change. This accounts for different starting masses. Show all working steps for full marks.

Active Transport

Active Transport: The movement of molecules or ions across a cell membrane from a region of lower concentration to higher concentration (AGAINST the concentration gradient), using energy (ATP) and carrier proteins

Comparison table:

Feature	Diffusion	Osmosis	Active Transport
Direction	Down gradient	Down gradient (water)	Against gradient
Energy needed?	No	No	Yes (ATP)
Carrier proteins?	Sometimes	Yes (aquaporins)	Yes (pumps)
Examples	O ₂ into blood	Water into roots	Mineral ions into roots; glucose reabsorption in kidney

1.4 Cell Division — Mitosis & Meiosis

Mitosis

Mitosis: A type of cell division producing TWO genetically identical daughter cells, each with the same number of chromosomes as the parent cell

Stages of Mitosis (IPMAT):

- **Interphase:** DNA replication occurs. Cell grows and prepares for division.
- **Prophase:** Chromosomes condense and become visible. Nuclear envelope breaks down.
- **Metaphase:** Chromosomes line up along the cell's equator (middle). Spindle fibres attach.
- **Anaphase:** Chromatids are pulled to opposite poles by spindle fibres.
- **Telophase:** Nuclear envelopes reform. Chromosomes uncoil. Cytoplasm divides (cytokinesis).

Uses of mitosis: Growth, repair of damaged tissues, asexual reproduction, replacement of worn-out cells (e.g. red blood cells, skin cells).

Meiosis

Meiosis: A type of cell division producing FOUR genetically different haploid daughter cells (gametes), each with HALF the number of chromosomes of the parent cell

Key features:

- Involves TWO divisions (Meiosis I and Meiosis II)
- Results in 4 haploid (n) cells from one diploid (2n) cell
- Genetic variation is introduced by: (1) crossing over and (2) independent assortment

- Occurs in the gonads (testes/ovaries)

Feature	Mitosis	Meiosis
Number of divisions	1	2
Daughter cells produced	2	4
Chromosome number	Diploid (2n) → Diploid (2n)	Diploid (2n) → Haploid (n)
Genetic variation?	No (identical)	Yes (varied)
Purpose	Growth, repair, replacement	Production of gametes (sex cells)

■ **EXAM TIP:** Learn these 5 differences by heart — Mitosis vs Meiosis is one of the most frequently examined topics in Paper 1 and Paper 2!

UNIT 2: HUMAN PHYSIOLOGY

2.1 Nutrition & Enzymes

Food Groups

A balanced diet contains all seven components in the right amounts:

Component	Role	Sources
Carbohydrates	Energy (respiration)	Rice, bread, pasta, glucose
Proteins	Growth, repair, enzymes, antibodies	Meat, fish, eggs, legumes
Lipids (Fats)	Energy store, cell membranes, thermal insulation	Butter, oil, nuts, avocado
Vitamins	Metabolic processes (varied)	Vitamin C (citrus), Vitamin D (sunlight)
Minerals	Bone, haemoglobin, enzyme cofactors	Calcium, Iron, Iodine
Fibre	Prevents constipation; aids peristalsis	Wholegrain, vegetables
Water	Solvent, transport, temperature regulation	Drinks, fruit, vegetables

Enzymes

Enzyme: A biological catalyst — a protein that speeds up chemical reactions without being used up

Active site: The specific region of an enzyme where the substrate binds, with a shape complementary to the substrate

Lock and Key Model: The substrate fits precisely into the active site of the enzyme, like a key in a lock

Factors affecting enzyme activity:

- **Temperature:** Rate increases up to the optimum temperature (~37°C in humans). Above optimum, the enzyme DENATURES — the active site changes shape permanently and the enzyme no longer functions.
- **pH:** Each enzyme has an optimum pH. Deviation causes denaturation. E.g. pepsin (stomach) optimum pH ≈ 2; amylase (saliva) optimum pH ≈ 7.
- **Substrate concentration:** Rate increases until all active sites are occupied (saturation point); after that, adding more substrate has no effect.
- **Enzyme concentration:** More enzyme molecules = more active sites available = faster rate (provided sufficient substrate).

WORKED EXAMPLE — Enzyme Experiment Analysis

An experiment measured amylase activity at different temperatures. Results showed fastest starch digestion at 37°C, with no digestion at 80°C. **Explain** these results.

At 37°C: this is the optimum temperature for human amylase. The enzyme-substrate complex forms most readily, so starch is broken down fastest. At 80°C: the high temperature breaks hydrogen bonds holding the enzyme's tertiary structure together. The active site changes shape and can no longer bind to starch. The enzyme is denatured (not dead — enzymes are not alive).

■ **EXAM TIP: NEVER say an enzyme is 'killed' by high temperature. Enzymes are not alive — they are DENATURED. The active site changes shape permanently.**

2.2 The Digestive System

Digestion: The breakdown of large, insoluble food molecules into small, soluble molecules that can be absorbed into the blood

Physical digestion: Breaking food into smaller pieces (chewing, churning in stomach) — increases surface area for enzymes.

Chemical digestion: Enzyme-catalysed hydrolysis of large molecules (starch → glucose, proteins → amino acids, lipids → fatty acids + glycerol).

Region	Key Enzyme	Reaction
Mouth	Amylase (salivary glands)	Starch → Maltose
Stomach	Pepsin	Proteins → Peptides (in acidic pH ≈ 2)
Small intestine	Amylase (pancreas), Maltase, Protease, Lipase	Starch→glucose; proteins→amino acids; lipids→fatty acids + glycerol
Large intestine	None	Water reabsorption; faeces formation

Absorption in the Small Intestine

The small intestine is highly adapted for absorption:

- Villi (finger-like projections) and microvilli increase surface area enormously
- Thin epithelial layer (one cell thick) → short diffusion distance
- Rich blood supply (capillaries) → maintains concentration gradient
- Lacteals (lymphatic vessels) absorb fatty acids and glycerol

■ **EXAM TIP: Questions about 'adaptations of the small intestine' appear almost every year. Always state the feature AND explain HOW it increases efficiency.**

2.3 The Respiratory System

Breathing (Ventilation): The physical process of moving air in and out of the lungs

Respiration: A chemical process occurring in ALL cells, releasing energy (ATP) from glucose

Aerobic respiration (with oxygen):

glucose + oxygen → carbon dioxide + water (+ATP energy)

Anaerobic respiration (without oxygen):

In animals: glucose → lactic acid (+small amount of ATP)

In yeast/plants: glucose → ethanol + carbon dioxide (+small amount of ATP)

Structure	Inhalation	Exhalation
Ribcage	Moves up and out	Moves down and in
Diaphragm	Contracts (flattens)	Relaxes (domes upward)
Volume of chest	Increases	Decreases
Air pressure in lungs	Decreases (below atmospheric)	Increases (above atmospheric)
Air movement	In (inhalation)	Out (exhalation)

Adaptations of the Alveoli for Gas Exchange

- Large total surface area (~70 m² in adult lungs) — many alveoli present
- Very thin walls (single cell layer, ~0.1 µm) — short diffusion distance
- Rich capillary network — maintains steep concentration gradient for O₂ and CO₂
- Moist lining — gases dissolve and diffuse more readily

■ **EXAM TIP: Gas exchange is PASSIVE (diffusion) — no energy needed. Breathing (ventilation) is ACTIVE — it requires energy (muscles of ribcage and diaphragm).**

2.4 The Circulatory System

Humans have a **double circulatory system**: blood passes through the heart twice for each complete circuit of the body.

- **Pulmonary circulation:** Heart → Lungs → Heart (oxygenation of blood)
- **Systemic circulation:** Heart → Body organs → Heart (delivery of O₂ and nutrients)

Vessel	Function	Features
Arteries	Carry blood AWAY from heart	Thick muscular wall, small lumen, no valves; blood under high pressure
Veins	Carry blood TOWARDS heart	Thin wall, large lumen, valves present; blood under low pressure
Capillaries	Exchange of materials between blood and tissues	One cell thick, very small, huge number and total surface area

Components of Blood

- **Red blood cells:** Transport oxygen (as oxyhaemoglobin). No nucleus. Biconcave.
- **White blood cells:** Immune defence. Phagocytes (engulf pathogens). Lymphocytes (produce antibodies).
- **Platelets:** Cell fragments involved in blood clotting.
- **Plasma:** Liquid portion; transports dissolved substances: CO₂, glucose, urea, hormones, proteins.

■ **EXAM TIP:** The heart has **FOUR** chambers: right atrium, right ventricle, left atrium, left ventricle. The **LEFT** side pumps oxygenated blood to the body and has a **THICKER** muscular wall than the right side.

2.5 Homeostasis

Homeostasis: The maintenance of a constant internal environment within the body, despite changes in external conditions

Thermoregulation — Body Temperature Control (37°C)

The hypothalamus in the brain acts as the thermostat, detecting blood temperature and coordinating responses.

Condition	Response	Explanation
Too hot	Vasodilation	Blood vessels near skin surface widen → more blood near surface → more heat lost by radiation
Too hot	Sweating	Water evaporates from skin surface → takes heat energy with it
Too hot	Hairs lie flat	Less insulating air trapped near skin
Too cold	Vasoconstriction	Blood vessels near surface narrow → less blood near surface → less heat lost
Too cold	Shivering	Rapid muscle contractions → generate heat by respiration
Too cold	Hairs stand up	More insulating air trapped near skin → reduces heat loss

Blood Glucose Regulation

Blood glucose is controlled by the pancreas through negative feedback:

- **Blood glucose too HIGH (after a meal):** Pancreas secretes **INSULIN** → liver/muscle cells take up glucose and convert it to glycogen (glycogenesis). Blood glucose falls back to normal.
- **Blood glucose too LOW (after exercise/fasting):** Pancreas secretes **GLUCAGON** → liver converts glycogen back to glucose (glycogenolysis) and releases it into blood. Blood glucose rises back to normal.

Negative feedback: A control mechanism in which the response reverses the original change, restoring the system to its set point

Diabetes mellitus Type 1: The pancreas does not produce enough insulin. Treated with insulin injections. Autoimmune condition.

Diabetes mellitus Type 2: Body cells do not respond properly to insulin (insulin resistance). Linked to obesity. Managed by diet, exercise, medication.

■ **EXAM TIP:** Diabetes questions often ask you to **EXPLAIN** why blood glucose remains high after a meal in a diabetic person. Answer: insufficient insulin is produced (Type 1) or cells do not respond to insulin (Type 2), so glucose is not taken up by liver/muscle cells.

2.6 The Nervous System & Hormones

Stimulus: A change in the environment (internal or external) detected by receptors

Response: The action taken by an effector as a result of a nervous or hormonal signal

Reflex arc pathway:

Receptor → Sensory neurone → Relay neurone (spinal cord) → Motor neurone → Effector

Reflex actions are RAPID, AUTOMATIC, and INVOLUNTARY responses that protect the body from harm (e.g. withdrawing hand from heat, blinking).

Nervous vs Hormonal Communication

Feature	Nervous Hormonal
Speed	Very fast (up to 120 m/s) Slow (via bloodstream)
Duration	Short-lived Long-lasting
Transmission	Electrical impulses along neurones Chemical messengers in blood
Target	Specific (one effector) General (many organs)
Examples	Reflex actions, voluntary movement Insulin, adrenaline, oestrogen

■ **EXAM TIP:** Adrenaline is the 'fight or flight' hormone. In exams, be ready to explain its effects: increased heart rate, dilated pupils, increased blood glucose, redirected blood to muscles.

2.7 Excretion — The Kidneys

Excretion: The removal of the metabolic waste products of chemical reactions in the body

Main excretory products:

- CO₂ and water — products of respiration, excreted via lungs
- Urea — product of amino acid deamination in liver; excreted by kidneys in urine

Kidney Function

The kidney produces urine in three stages:

1. **Ultrafiltration** (glomerulus/Bowman's capsule): High pressure forces small molecules (glucose, urea, water, ions) out of blood into the nephron. Blood cells and large proteins stay in blood.
2. **Selective reabsorption** (proximal tubule): ALL glucose and amino acids, most water and ions are reabsorbed back into the blood. Urea is NOT reabsorbed (remains in urine).
3. **Osmoregulation** (loop of Henle / collecting duct): ADH hormone controls water reabsorption. More ADH → more water reabsorbed → concentrated urine produced.

■ **EXAM TIP:** Glucose should NEVER be found in normal urine. If glucose is detected in a urine test, this indicates diabetes mellitus — blood glucose was so high that not all glucose could be reabsorbed.

UNIT 3: BASIC ECOLOGY

3.1 Ecosystems, Habitats & Niches

Ecosystem: All the living organisms in an area together with their non-living (abiotic) environment, interacting as a system

Habitat: The place where an organism lives (e.g. oak tree, pond, rainforest floor)

Population: All the organisms of ONE species living in a defined area at one time

Community: All the populations of DIFFERENT species living in an area at one time

Abiotic factors: Non-living components: temperature, light intensity, rainfall, pH, salinity, wind speed

Biotic factors: Living components: predation, competition, disease, food availability, pollination

Niche: The role of an organism in its ecosystem — includes what it eats, where it lives, and how it interacts with other organisms. No two species occupy exactly the same niche.

3.2 Food Chains & Food Webs

A food chain shows the flow of ENERGY through organisms. Arrows mean 'energy passes to' or 'is eaten by'.

Example: Grass → Grasshopper → Frog → Hawk

Trophic levels:

- **Producers (Trophic level 1):** Photosynthetic organisms (green plants, algae). Convert light energy into chemical energy.
- **Primary consumers (TL2):** Herbivores. Eat producers (e.g. grasshopper).
- **Secondary consumers (TL3):** Carnivores. Eat primary consumers (e.g. frog).
- **Tertiary consumers (TL4):** Eat secondary consumers (e.g. hawk).
- **Decomposers:** Bacteria and fungi. Break down dead organic matter, returning nutrients to soil.

■ WORKED EXAMPLE — Food Web Analysis (Cambridge-style)

A food web contains: Oak tree → Caterpillar → Blue tit → Sparrowhawk; Oak tree → Squirrel → Sparrowhawk; Oak tree → Aphid → Blue tit.

Question: If the sparrowhawk population decreases, predict and explain the effect on the caterpillar population.

The caterpillar population would INCREASE. With fewer sparrowhawks, the blue tit population increases (less predation). Increased blue tit numbers would cause caterpillar numbers to DECREASE. However, since blue tits also eat aphids, aphid numbers could increase, which in turn could reduce the oak tree's resources.

■ **EXAM TIP:** Food web questions ask for PREDICTIONS with explanations. Always state (1) the immediate effect, then (2) the knock-on effects through the food web. Use specific population names, not just 'prey' or 'predator'.

3.3 Energy Flow & Ecological Pyramids

Energy is lost at each trophic level — it is not recycled. This is why food chains rarely have more than 4–5 trophic levels.

How energy is lost between trophic levels:

- Heat energy lost through respiration (~60–70% of energy is used for cellular respiration)
- Undigested material passes out as faeces/urine (lost via egestion/excretion)
- Energy used for growth, movement, reproduction

Efficiency of energy transfer: Approximately 10% of energy passes from one trophic level to the next. So only 10% of energy in producers reaches primary consumers, and 1% reaches secondary consumers.

WORKED EXAMPLE — Energy Transfer Calculation

A field contains 100,000 kJ of energy in grass (producers). Assuming 10% efficiency at each trophic level:

- Energy in primary consumers (rabbits) = $100,000 \times 10\% = \mathbf{10,000 \text{ kJ}}$
- Energy in secondary consumers (foxes) = $10,000 \times 10\% = \mathbf{1,000 \text{ kJ}}$
- Energy in tertiary consumers = $1,000 \times 10\% = \mathbf{100 \text{ kJ}}$

Ecological Pyramids:

- **Pyramid of numbers:** Number of organisms at each trophic level. Can be inverted (e.g. one oak tree supports thousands of caterpillars).
- **Pyramid of biomass:** Mass of organisms at each level. Usually pyramid-shaped. Rarely inverted.
- **Pyramid of energy:** Energy at each trophic level. ALWAYS pyramid-shaped — never inverted. Most reliable measure.

EXAM TIP: A pyramid of energy is ALWAYS a true pyramid (wider at base, narrower at top). If asked to draw one, label axes with 'trophic level' (y-axis) and 'energy / kJ per year' (x-axis).

3.4 Nutrient Cycles

The Carbon Cycle

Carbon is constantly cycled between living organisms and the atmosphere:

- **Photosynthesis:** Plants absorb CO_2 from atmosphere → converted to organic compounds
- **Respiration:** All organisms release CO_2 back into atmosphere
- **Feeding:** Carbon passes from plants to consumers along food chains
- **Decomposition:** Decomposers break down dead matter → CO_2 released
- **Combustion:** Burning fossil fuels and wood releases stored carbon as CO_2
- **Fossilisation:** Dead organisms form coal, oil, gas over millions of years (carbon store)

The Nitrogen Cycle

Nitrogen is essential for making proteins and nucleic acids (DNA). Atmospheric N_2 cannot be used directly by most organisms.

- **Nitrogen fixation:** Nitrogen-fixing bacteria (e.g. Rhizobium in root nodules) convert N_2 → ammonia (NH_3) / ammonium ions
- **Nitrification:** Nitrifying bacteria convert ammonium → nitrites → nitrates (absorbed by plants)
- **Assimilation:** Plants absorb nitrates → make amino acids → proteins → eaten by consumers
- **Ammonification:** Decomposers break down proteins in dead matter → ammonia released
- **Denitrification:** Denitrifying bacteria convert nitrates back to N_2 gas (waterlogged soils)

■ **EXAM TIP:** Examiners love to ask about the role of **BACTERIA** in the nitrogen cycle. Name the type and state its specific function. **Rhizobium** (nitrogen fixation), **Nitrosomonas/Nitrobacter** (nitrification), **denitrifying bacteria** (denitrification).

3.5 Population Dynamics

Population growth: Change in population size over time, affected by birth rate, death rate, immigration, and emigration

S-shaped (sigmoid) growth curve — typical population growth:

- **Lag phase:** Slow growth — population small, adapting to environment
- **Log (exponential) phase:** Rapid growth — resources plentiful, few predators
- **Stationary phase:** Population stabilises at carrying capacity (K)

Carrying capacity (K): The maximum population size that an environment can sustain indefinitely with available resources

Limiting factors that prevent unlimited growth:

- Food/water shortage (competition for resources)
- Increased predation (more predators at higher prey densities)
- Disease (spreads more rapidly in dense populations)
- Accumulation of waste products

■ WORKED EXAMPLE — Predator-Prey Relationships

A study of Canadian lynx and snowshoe hare over 90 years shows a cyclic relationship. Explain why hare populations peak BEFORE lynx populations.

When hare numbers increase, there is more food for lynx. The lynx population increases, but with a time lag (it takes time to breed). The increased lynx predation then causes hare numbers to fall. With less prey, lynx numbers also fall — again with a time lag. The cycle then repeats. This is a classic example of predator-prey cycling.

3.6 Human Impacts & Conservation

Human Activity	Environmental Impact
Deforestation	Loss of biodiversity, habitat destruction, increased CO ₂ (less photosynthesis), soil erosion, disruption of water cycle
Pollution (water)	Eutrophication: fertiliser runoff → algal bloom → algae die → decomposers respire → O ₂ depleted → fish die
Pollution (air)	Acid rain (SO ₂ , NO ₂) damages ecosystems; global warming (CO ₂ , CH ₄) → climate change
Overfishing	Reduced fish populations; trophic cascade effects; potential extinction of species
Monoculture farming	Reduced biodiversity, increased pesticide/herbicide use, soil degradation

Conservation Strategies

- **Protected areas:** National parks, nature reserves — preserve habitats
- **Captive breeding programmes:** Breed endangered species in zoos → reintroduce to wild
- **Seed banks:** Store seeds of endangered plants (e.g. Svalbard Global Seed Vault)
- **Legislation:** CITES (international trade in endangered species), fishing quotas
- **Sustainable forestry:** Replant trees at same rate as felling

■ **EXAM TIP:** Eutrophication questions are common! The sequence is: fertiliser → leaching into water → algal bloom → algae die → decomposers increase → aerobic respiration by decomposers depletes O₂ → fish and other organisms die due to lack of O₂.

PRACTICE QUESTIONS

Section A — Short Answer Questions (1–2 marks each)

Attempt these questions before checking the answers. These mirror Paper 1 and Paper 2 style questions.

Q1. State TWO differences between mitosis and meiosis.

Answer: Any two from: (1) Mitosis produces 2 cells; meiosis produces 4. (2) Mitosis produces diploid cells; meiosis produces haploid cells. (3) Daughter cells of mitosis are genetically identical; meiosis produces genetic variation. (4) Mitosis involves one division; meiosis involves two.

Q2. Define osmosis.

Answer: The net movement of water molecules from a region of higher water potential to a region of lower water potential, through a selectively permeable membrane.

Q3. Name the organelle where aerobic respiration occurs.

Answer: Mitochondria (mitochondrion).

Q4. State the word equation for aerobic respiration.

Answer: Glucose + oxygen → carbon dioxide + water (+ energy / ATP).

Q5. Give TWO ways in which a red blood cell is adapted for transporting oxygen.

Answer: Any two: (1) Biconcave disc shape increases surface area. (2) No nucleus — more room for haemoglobin. (3) Contains large amounts of haemoglobin to bind O₂. (4) Thin flexible membrane allows it to squeeze through capillaries.

Q6. Explain why body temperature is maintained at 37°C.

Answer: Enzymes in the body work at their optimum rate at 37°C; higher temperatures would cause denaturation (change in active site shape), reducing enzyme activity and metabolic rate.

Q7. What is the role of insulin in blood glucose regulation?

Answer: Insulin (secreted by the pancreas) causes liver and muscle cells to take up glucose from the blood and convert it to glycogen, lowering blood glucose concentration back to normal.

Q8. Explain what is meant by 'carrying capacity'.

Answer: The maximum population size that an environment can sustainably support with available food, water, space, and other resources.

Q9. Name TWO roles of bacteria in the nitrogen cycle.

Answer: Any two: nitrogen fixation (converting N₂ to ammonia); nitrification (ammonia to nitrates); denitrification (nitrates to N₂); decomposition/ammonification.

Q10. Explain how eutrophication leads to the death of fish.

Answer: Fertiliser runoff → algal bloom on water surface → algae block light → aquatic plants die → decomposers break down dead matter → decomposers use up O₂ by aerobic respiration → O₂ levels drop → fish die due to lack of O₂.

Section B — Structured Questions (4–6 marks each)

These questions require extended answers. Each mark point is listed in the model answer.

Q11. A student investigates the effect of temperature on the enzyme amylase. She measures the time for starch to be fully digested at 10°C, 20°C, 37°C, 60°C, and 80°C. (a) Predict the results. (b) Explain the result at 37°C. (c) Explain the result at 80°C. [6 marks]

Answer: (a) Fastest digestion (shortest time) at 37°C; increasing rate from 10°C to 37°C; little or no digestion at 80°C. [2] (b) 37°C is the optimum temperature for amylase; maximum enzyme-substrate (amylase-starch) collisions per second; active site shape perfectly complementary to substrate. [2] (c) 80°C is above optimum; excessive heat breaks hydrogen bonds in enzyme tertiary structure; active site changes shape; substrate can no longer fit/bind; enzyme is denatured; reaction stops. [2]

Q12. Describe the process of selective reabsorption in the kidney. [4 marks]

Answer: Filtered fluid (filtrate) flows along the proximal convoluted tubule [1]; ALL glucose and amino acids are reabsorbed into surrounding capillaries by active transport (against concentration gradient, requires ATP) [1]; most water is reabsorbed by osmosis [1]; most ions reabsorbed; urea is NOT reabsorbed and remains in the tubule to be excreted [1].

Q13. Explain how deforestation can lead to increased atmospheric CO₂ and reduced biodiversity. [5 marks]

Answer: Burning trees releases CO₂ by combustion [1]; loss of trees means less photosynthesis, so less CO₂ is absorbed from atmosphere [1]; decomposition of dead organic matter also releases CO₂ [1]. Reduced biodiversity: loss of habitat destroys food webs and niches [1]; many species that depend on the forest for shelter, food, or breeding sites are lost (extinction or local extinction) [1].

Section C — Extended Writing Questions (6+ marks)

Write in continuous prose. These questions assess your ability to apply knowledge and evaluate information.

Q14. Explain how the human body maintains a constant core body temperature of 37°C when a person moves from a cold environment into a warm one. [6 marks]

Answer: When in a cold environment: thermoreceptors in skin and hypothalamus detect low temperature; vasoconstriction occurs (blood vessels near skin surface narrow) reducing blood flow to skin, decreasing heat loss by radiation; hairs erect trapping insulating air layer; shivering — rapid muscle contractions — generates heat through respiration [3 marks for cold responses]. When moving into warm environment: hypothalamus detects rise in blood temperature; vasodilation — blood vessels near skin dilate, more blood flows near surface, heat lost by radiation; sweating — sweat glands secrete water onto skin, evaporation requires latent heat of vaporisation absorbed from skin, cooling body; hairs flatten — less air trapped, less insulation [3 marks for warm responses]. Negative feedback: as body temperature returns to 37°C (set point), the corrective responses are switched off.

Essential Definitions Checklist

Use this checklist before your exam. Tick each definition once you can state it accurately without notes:

- Cell (basic unit of life)
- Diffusion (net movement, down gradient, random movement)
- Osmosis (water, semi-permeable membrane, water potential)
- Active transport (against gradient, energy, carrier proteins)
- Enzyme (biological catalyst, protein, active site)
- Mitosis (2 genetically identical diploid daughter cells)
- Meiosis (4 genetically different haploid gametes)

- Homeostasis (constant internal environment)
- Negative feedback (response reverses change)
- Respiration (chemical process, energy/ATP from glucose)
- Photosynthesis (light energy → chemical energy, $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{glucose} + \text{O}_2$)
- Ecosystem (organisms + abiotic environment, interacting)
- Niche (role of an organism in its ecosystem)
- Carrying capacity (maximum sustainable population size)
- Eutrophication (fertiliser → algal bloom → O_2 depletion → death)

Good luck in your IGCSE Biology examination!

Remember: Read each question carefully, use technical terminology, and always show your reasoning.